Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-16 SWL Wide Berm 2% - Test 1.experiment 60% of (4) use*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.336 Hz*
Freq. upper bound:	*6.719 Hz*

Freq. Domain Analysis

Overlap of (4) timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*26*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*29*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *25.3%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.08257*	*0.0209*
Peak wave period T_P (s):	*1.488*	*1.6*
Energy wave period $T_{-1,0}$ (s):	*1.444*	*1.375*
Mean wave period $T_{0,1}$ (s):	*1.368*	*1.156*
Mean wave period $T_{0,2}$ (s):	*1.318*	*1.026*
Spectral width (Broadness):	*0.5951*	*0.793*
Spectral width (Narrowness):	*0.2766*	*0.5176*

Time Domain Analysis

Overall amplitude reflection coefficient: *25.4%*

	Incident	**Reflected**
Number of waves:	1001	1228
Mean wave height H_m (m):	*0.05242*	*0.01254*
Mean wave period T_m (s):	*1.366*	*1.113*
Sig. wave height H_s (m):	*0.08046*	*0.02091*
$T_{H_{1/3}}$ (s):	*1.467*	*1.387*
H_{max} (m):	*0.1344*	*0.04329*
T_{Hmax} (s):	*1.63*	*1.367*
$H_{1/10}$ (m):	*0.09831*	*0.0276*
$H_{1/50}$ (m):	*0.1153*	*0.03432*
$H_{1/100}$ (m)	*0.1235*	*0.0365*
$H_{1/250}$ (m):	*0.1316*	*0.04006*
$H_{0.1\%}$ (m):	*0.1344*	*0.04267*
$H_{1\%}$ (m):	*0.1149*	*0.03255*
$H_{2\%}$ (m):	*0.1035*	*0.03141*
$H_{10\%}$ (m)	*0.08651*	*0.0229*
Groupiness factor GF:	*0.9664*	*1.202*
Skewness b1:	*0.3357*	*0.1897*
Kurtosis b2:	*2.915*	*3.686*

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Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-16 SWL Wide Berm 2% - Test 1.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.266 Hz*
Freq. upper bound:	*5.313 Hz*

Freq. Domain Analysis

Overlap of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*29*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*36*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *30%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1068*	*0.03207*
Peak wave period T_P (s):	*1.73*	*1.882*
Energy wave period $T_{-1,0}$ (s):	*1.618*	*1.589*
Mean wave period $T_{0,1}$ (s):	*1.491*	*1.24*
Mean wave period $T_{0,2}$ (s):	*1.405*	*1.046*
Spectral width (Broadness):	*0.716*	*0.8407*
Spectral width (Narrowness):	*0.3538*	*0.6364*

Time Domain Analysis

Overall amplitude reflection coefficient: *30.1%*

	Incident	**Reflected**
Number of waves:	1005	1123
Mean wave height H_m (m):	*0.06779*	*0.02035*
Mean wave period T_m (s):	*1.518*	*1.359*
Sig. wave height H_s (m):	*0.1055*	*0.0336*
$T_{H_{1/3}}$ (s):	*1.697*	*1.694*
H_{max} (m):	*0.1874*	*0.0635*
T_{Hmax} (s):	*1.659*	*1.71*
$H_{1/10}$ (m):	*0.1315*	*0.04421*
$H_{1/50}$ (m):	*0.161*	*0.05409*
$H_{1/100}$ (m)	*0.1723*	*0.05719*
$H_{1/250}$ (m):	*0.1819*	*0.0603*
$H_{0.1\%}$ (m):	*0.1874*	*0.0631*
$H_{1\%}$ (m):	*0.1548*	*0.05392*
$H_{2\%}$ (m):	*0.1466*	*0.04835*
$H_{10\%}$ (m)	*0.112*	*0.03658*
Groupiness factor GF:	*1.074*	*1.251*
Skewness b1:	*0.5602*	*0.2762*
Kurtosis b2:	*3.293*	*3.621*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-16 SWL Wide Berm 2% - Test 1.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.250 Hz*
Freq. upper bound:	*5.000 Hz*

	Gauge 1	Gauge 2	Gauge 3
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Freq. Domain Analysis

Overlap of spectral series:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*32*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*38*
Hilbert filter length:	*2048*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *31.8%*

	Incident	Reflected
Sig. wave height H_{m0} (m):	*0.1218*	*0.0387*
Peak wave period T_P (s):	*2*	*2*
Energy wave period $T_{-1,0}$ (s):	*1.766*	*1.736*
Mean wave period $T_{0,1}$ (s):	*1.58*	*1.257*
Mean wave period $T_{0,2}$ (s):	*1.457*	*1.021*
Spectral width (Broadness):	*0.7816*	*0.8579*
Spectral width (Narrowness):	*0.4188*	*0.7191*

Time Domain Analysis

Overall amplitude reflection coefficient: *31.8%*

	Incident	Reflected
Number of waves:	997	1143
Mean wave height H_m (m):	*0.07877*	*0.02481*
Mean wave period T_m (s):	*1.676*	*1.463*
Sig. wave height H_s (m):	*0.1231*	*0.0405*
$T_{H_{1/3}}$ (s):	*1.881*	*1.816*
H_{max} (m):	*0.2109*	*0.09333*
T_{Hmax} (s):	*1.76*	*1.13*
$H_{1/10}$ (m):	*0.1562*	*0.05313*
$H_{1/50}$ (m):	*0.1901*	*0.06751*
$H_{1/100}$ (m)	*0.2003*	*0.07407*
$H_{1/250}$ (m):	*0.2057*	*0.08318*
$H_{0.1\%}$ (m):	*Not enough data*	*0.09307*
$H_{1\%}$ (m):	*0.1932*	*0.06574*
$H_{2\%}$ (m):	*0.1752*	*0.05864*
$H_{10\%}$ (m)	*0.1312*	*0.04259*
Groupiness factor GF:	*1.123*	*1.255*
Skewness b1:	*0.7408*	*0.1769*
Kurtosis b2:	*3.571*	*3.599*

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Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-16 SWL Wide Berm 2% - Test 1.experiment_1.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth:	*0.365 m*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.242 Hz*
Freq. upper bound:	*4.844 Hz*

Freq. Domain Analysis

Overlap of (4) series:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*34*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*40*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *32.8%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1321*	*0.04329*
Peak wave period T_P (s):	*2.065*	*2.065*
Energy wave period $T_{-1,0}$ (s):	*1.83*	*1.803*
Mean wave period $T_{0,1}$ (s):	*1.613*	*1.253*
Mean wave period $T_{0,2}$ (s):	*1.469*	*1.001*
Spectral width (Broadness):	*0.8122*	*0.8633*
Spectral width (Narrowness):	*0.4524*	*0.7518*

Time Domain Analysis

Overall amplitude reflection coefficient: *32.8%*

	Incident	**Reflected**
Number of waves:	1018	1191
Mean wave height H_m (m):	*0.0872*	*0.02872*
Mean wave period T_m (s):	*1.758*	*1.504*
Sig. wave height H_s (m):	*0.1345*	*0.04574*
$T_{H_{1/3}}$ (s):	*1.975*	*1.833*
H_{max} (m):	*0.2306*	*0.08734*
T_{Hmax} (s):	*1.871*	*1.066*
$H_{1/10}$ (m):	*0.1671*	*0.05851*
$H_{1/50}$ (m):	*0.1949*	*0.07218*
$H_{1/100}$ (m)	*0.2033*	*0.07651*
$H_{1/250}$ (m):	*0.2136*	*0.08087*
$H_{0.1\%}$ (m):	*0.2303*	*0.08689*
$H_{1\%}$ (m):	*0.1935*	*0.071*
$H_{2\%}$ (m):	*0.1828*	*0.06478*
$H_{10\%}$ (m)	*0.1452*	*0.04909*
Groupiness factor GF:	*1.085*	*1.162*
Skewness b1:	*0.7877*	*0.1694*
Kurtosis b2:	*3.477*	*3.401*

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Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-16 SWL Wide Berm 2% - Test 1.experiment_1.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth:	*0.365 m*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.227 Hz*
Freq. upper bound:	*4.531 Hz*

Freq. Domain Analysis

Overlap of (4) series:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*35*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*42*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *34.5%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1437*	*0.04961*
Peak wave period T_P (s):	*2.207*	*2.207*
Energy wave period $T_{-1,0}$ (s):	*1.886*	*1.872*
Mean wave period $T_{0,1}$ (s):	*1.628*	*1.229*
Mean wave period $T_{0,2}$ (s):	*1.465*	*0.971*
Spectral width (Broadness):	*0.829*	*0.856*
Spectral width (Narrowness):	*0.485*	*0.776*

Time Domain Analysis

Overall amplitude reflection coefficient: *34.6%*

	Incident	**Reflected**
Number of waves:	1024	1230
Mean wave height H_m (m):	*0.09537*	*0.03321*
Mean wave period T_m (s):	*1.804*	*1.502*
Sig. wave height H_s (m):	*0.147*	*0.05238*
$T_{H_{1/3}}$ (s):	*2.062*	*1.775*
H_{max} (m):	*0.2255*	*0.09443*
T_{Hmax} (s):	*1.901*	*1.791*
$H_{1/10}$ (m):	*0.18*	*0.06578*
$H_{1/50}$ (m):	*0.1994*	*0.07954*
$H_{1/100}$ (m)	*0.2066*	*0.0843*
$H_{1/250}$ (m):	*0.2164*	*0.08953*
$H_{0.1\%}$ (m):	*0.2255*	*0.09331*
$H_{1\%}$ (m):	*0.1979*	*0.07751*
$H_{2\%}$ (m):	*0.1901*	*0.0716*
$H_{10\%}$ (m)	*0.1624*	*0.05674*
Groupiness factor GF:	*1.073*	*1.114*
Skewness b1:	*0.8369*	*0.1437*
Kurtosis b2:	*3.452*	*3.264*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-16 SWL Wide Berm 2% - Test 1.experiment_1.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth:	*0.365 m*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.219 Hz*
Freq. upper bound:	*4.375 Hz*

Freq. Domain Analysis

Overlapping of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*37*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*44*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *35.8%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1534*	*0.05484*
Peak wave period T_P (s):	*2.286*	*2.286*
Energy wave period $T_{-1,0}$ (s):	*1.918*	*1.895*
Mean wave period $T_{0,1}$ (s):	*1.627*	*1.214*
Mean wave period $T_{0,2}$ (s):	*1.449*	*0.954*
Spectral width (Broadness):	*0.8373*	*0.8535*
Spectral width (Narrowness):	*0.5107*	*0.7864*

Time Domain Analysis

Overall amplitude reflection coefficient: *35.8%*

	Incident	**Reflected**
Number of waves:	1036	1234
Mean wave height H_m (m):	*0.1035*	*0.03772*
Mean wave period T_m (s):	*1.836*	*1.542*
Sig. wave height H_s (m):	*0.158*	*0.05783*
$T_{H_{1/3}}$ (s):	*2.104*	*1.808*
H_{max} (m):	*0.238*	*0.0927*
T_{Hmax} (s):	*2.414*	*1.643*
$H_{1/10}$ (m):	*0.1885*	*0.07057*
$H_{1/50}$ (m):	*0.2104*	*0.08175*
$H_{1/100}$ (m)	*0.2184*	*0.08565*
$H_{1/250}$ (m):	*0.2278*	*0.09032*
$H_{0.1\%}$ (m):	*0.2376*	*0.09233*
$H_{1\%}$ (m):	*0.2086*	*0.07925*
$H_{2\%}$ (m):	*0.1989*	*0.07596*
$H_{10\%}$ (m)	*0.1695*	*0.0619*
Groupiness factor GF:	*1.069*	*1.054*
Skewness b1:	*0.9135*	*0.1082*
Kurtosis b2:	*3.5*	*3.063*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-16 SWL Wide Berm 2% - Test 1.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth:	*0.365 m*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.219 Hz*
Freq. upper bound:	*4.375 Hz*

Freq. Domain Analysis

Overlap of segments:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*38*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*44*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *36.8%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1578*	*0.0581*
Peak wave period T_P (s):	*2.286*	*2.56*
Energy wave period $T_{-1,0}$ (s):	*1.953*	*1.912*
Mean wave period $T_{0,1}$ (s):	*1.636*	*1.198*
Mean wave period $T_{0,2}$ (s):	*1.445*	*0.9382*
Spectral width (Broadness):	*0.8481*	*0.8526*
Spectral width (Narrowness):	*0.5308*	*0.7948*

Time Domain Analysis

Overall amplitude reflection coefficient: *37%*

	Incident	**Reflected**
Number of waves:	1039	1286
Mean wave height H_m (m):	*0.1087*	*0.0404*
Mean wave period T_m (s):	*1.891*	*1.529*
Sig. wave height H_s (m):	*0.1638*	*0.06092*
$T_{H_{1/3}}$ (s):	*2.18*	*1.858*
H_{max} (m):	*0.2438*	*0.09643*
T_{Hmax} (s):	*1.626*	*2.179*
$H_{1/10}$ (m):	*0.1931*	*0.0731*
$H_{1/50}$ (m):	*0.2189*	*0.08672*
$H_{1/100}$ (m)	*0.2287*	*0.09168*
$H_{1/250}$ (m):	*0.2375*	*0.09523*
$H_{0.1\%}$ (m):	*0.2435*	*0.0962*
$H_{1\%}$ (m):	*0.217*	*0.08506*
$H_{2\%}$ (m):	*0.2049*	*0.0793*
$H_{10\%}$ (m)	*0.1743*	*0.06456*
Groupiness factor GF:	*1.053*	*1.004*
Skewness b1:	*0.9364*	*0.06366*
Kurtosis b2:	*3.457*	*2.912*

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Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-16 SWL Wide Berm 2% - Test 1.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth:	*0.365 m*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.211 Hz*
Freq. upper bound:	*4.219 Hz*

Freq. Domain Analysis

Overlapping of (4) series:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*39*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*46*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *37.7%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1656*	*0.06237*
Peak wave period T_P (s):	*2.37*	*2.462*
Energy wave period $T_{-1,0}$ (s):	*1.986*	*1.953*
Mean wave period $T_{0,1}$ (s):	*1.645*	*1.206*
Mean wave period $T_{0,2}$ (s):	*1.446*	*0.9429*
Spectral width (Broadness):	*0.8477*	*0.8479*
Spectral width (Narrowness):	*0.5419*	*0.7979*

Time Domain Analysis

Overall amplitude reflection coefficient: *37.7%*

	Incident	**Reflected**
Number of waves:	1032	1315
Mean wave height H_m (m):	*0.1164*	*0.04355*
Mean wave period T_m (s):	*1.953*	*1.532*
Sig. wave height H_s (m):	*0.1713*	*0.06304*
$T_{H_{1/3}}$ (s):	*2.246*	*1.833*
H_{max} (m):	*0.2362*	*0.1179*
T_{Hmax} (s):	*2.149*	*2.287*
$H_{1/10}$ (m):	*0.1969*	*0.07608*
$H_{1/50}$ (m):	*0.217*	*0.09005*
$H_{1/100}$ (m)	*0.2248*	*0.09588*
$H_{1/250}$ (m):	*0.2317*	*0.1034*
$H_{0.1\%}$ (m):	*0.2362*	*0.1135*
$H_{1\%}$ (m):	*0.2146*	*0.08826*
$H_{2\%}$ (m):	*0.205*	*0.08212*
$H_{10\%}$ (m)	*0.1821*	*0.06619*
Groupiness factor GF:	*1.04*	*0.9862*
Skewness b1:	*0.9831*	*0.03908*
Kurtosis b2:	*3.429*	*2.861*

Department of Civil Engineering

Aalborg University

Project title	
Drawn by	
Date	*16. mrt 2026*
Remarks	
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